

The effect of irradiated PTFE on the friction and wear behavior of chemically bonded PA46-PTFE-cb and PA66-PTFE-cb compounds

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ABSTRACT

In this paper, various dry lubricants made of a polyamide (PA) matrix and two different grades of radiation-modified PTFE, which are chemically bonded to the matrix by reactive melt extrusion, are examined regarding their tribological suitability and compared to a PA-PTFE reference compound with non-modified PTFE. For this purpose, a block-on-ring test rig was used under ambient conditions, where the block made of dry lubricant is sliding on a steel ring made of 16MnCr5 with constant sliding speed. Results of tribological tests on the block-on-ring test rig such as wear and friction depending on the contact pressure are presented. In addition to that, analyses of the layer formation of the lubricants on steel counterparts are shown. An essential finding of the investigations is that the mass thickness on the tribologically stressed surfaces to be lubricated with the dry lubricant is not directly related to the compound wear, but rather depends on the pressure in the steel-compound contact. Furthermore, examinations showed that an increase of pressure in the block-ring contact does not lead to an increase of the amount of dry lubricant material on the steel surface.

1. Introduction

In transmission applications, oil or grease are usually used as lubricants for machine elements such as bearings or gears. In some applications, these types of lubricants reach their limits with regard to their applicability. For instance, in the food industry or in the medical sector, where clean rooms are required, the use of oil and grease is not suitable in some cases. Also, for applications under extreme temperature conditions, fluid lubricants no longer fulfill their purpose. Therefore, solid lubricants are often used in such applications, where compounds based on e.g. polytetrafluoroethylene (PTFE) are common options.

PTFE is characterized by a wide operating temperature range ($-250 \dots +270$ °C), high chemical resistance and excellent non-stick properties, which leads to the low friction coefficients and moderate wear rates ($k \approx 10^{-5} \text{ mm}^3/\text{Nm}$ at sliding speeds well below 5 mm/s) in contact with steel counter sample. Therefore, various studies have shown that adding a different material, namely, filler, in PTFE can improve its wear resistance. Khare et al. investigated the formation of PTFE transfer films on

steel surfaces [1], using PTFE nanocomposite materials with alpha phase aluminum oxide nanoparticles. After a running-in phase, the PTFE is transferred to the mating body, which leads to stable friction and wear behavior. Surface analysis and molecular dynamics simulation have shown that, in addition to defluorination, carbonyl and hydroxyl groups are also formed during transfer film formation [2]. Li et al. showed that low speeds and loads as well as an isotropic surface increase the transfer film formation of PTFE/copper composites on a GCr15 counter surface [3]. On surfaces where the direction of movement was perpendicular to the machining direction, the most stable and largest transfer films could be observed up to a film thickness of 5 μm .

Compared to PTFE, polyamides (PA) are characterized by high strength, high wear resistance, high friction coefficients in relative movement with steel counterparts and low temperature-dependent deformation, among other things. Due to their relatively high modulus of elasticity, they can be used at high loads and high temperatures (up to 100 °C) [4]. The high friction coefficient of PA makes it however less suitable for use in tribological systems as dry lubricants. Various studies

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have shown that the combination of polyamides with solid lubricants, inorganic particles, or fibers can lead to an improvement in their tribological properties [5–7]. The influence of glass fiber reinforcement and MoS₂ lubricant on the wear and temperature behavior of journal bearings made of PA66 was investigated by Demirci et al. [8]. Shin et al. investigated the influence of glass fiber on the tribological properties of PA66 in a block-on-ring test configuration. It was seen that the friction coefficient and wear rate of the compound decreased with increasing volume fraction of the glass fiber [4,9].

Compared to other solid lubricants such as graphite or MoS₂, PTFE is one of the most effective solid lubricants, so it is used in a wide range of applications [10]. However, physical blends of PA-PTFE show inferior concerning mechanical properties compared to PA due to incompatibility between the polar polyamide matrix and the non-polar PTFE filler. Improving the polarity of PTFE, which can be achieved by electron beam irradiation, is one of the solutions for this incompatibility [11].

Although PTFE is one of the most chemically and thermally stable plastic materials, PTFE shows a low resistance to high-energy radiation. When irradiated in the presence of air, the C–C and C–F bonds of PTFE are broken, resulting in micro-powders with lower molecular weight, functional groups (–COF and –COOH) and persistent perfluoroalkyl (peroxy) radicals [12,13]. The hydrophilic functional groups can be used to create a chemically covalent bond to other monomers/polymers (e.g. polyamides). Franke et al. investigated the tribological behavior of chemically bonded (in the following abbreviated as cb) PTFE-polyamide compounds [14]. It was shown that with a mass fraction of PTFE between 3.3 and 50 wt-%, the friction and wear coefficient of the polyamides PA6, PA66 and PA12 can be improved under dry friction. The study of Muraki et al. shows that PA's friction and wear characteristics have been improved by blending 10 wt-% PTFE into the PA matrix, whereby the chemically blended material PA/PTFE outperforms the physically blended material PA/PTFE [15]. The influence of irradiated PTFE on the properties of PA6/PTFE blends was also investigated by Liu et al. It was shown, that the PTFE modified by ϵ -caprolactam could greatly improve the abrasive resistance of PA6 as well as the coefficient of friction [16].

In [17,18], the chemical bonding of PA66-PTFE-oil-cb compounds, which were produced via a two-step reactive extrusion process was shown. It was proven that additional chemically bonded oil molecules result in a further significantly lowering of the coefficient of friction in block-on-ring measurements compared to PA66-PTFE-cb and plain PA66 respectively [17,18]. In this article, research on the transfer mechanism of compound materials based on PA and radiation-modified PTFE as lubricant to steel counter surfaces is presented. For this purpose, a block-on-ring test arrangement is used, with blocks made of different PA-PTFE-cb compounds as dry lubricant and the ring made of steel. The compounds used vary in terms of the matrix polymer and the PTFE micro powder used. The tribological tests, in which wear and also friction is evaluated, are concluded by surface characterization to determine the amount of compound transferred to the steel surface.

The background of the presented work is the goal of lubricating the tooth contact in worm gears with the material pairing steel-bronze with the help of solid lubricants. Investigations on the transfer mechanism of the solid lubricant to the steel surface form the basis for the constructive design of the tribological system of a solid lubricated worm gear. Similar studies have been conducted on gear transmissions, where a sacrificial gear consisting of the solid lubricant is designed to lubricate the tooth contacts of the power transmitting gear pair [19].

2. Experimental method

2.1. Materials and processing

In this study, three different commercially available PTFE micro powders were used to couple with PA46 Stanyl TW300 (DSM, Grenzach-

Wyhlen, Germany), and PA66 (Technyl A205F Solvay, Hannover, Germany). The used PTFE types and their properties are listed in Table 1. All PTFE-Types were produced by Chemours, Wilmington, Delaware, USA.

Due to the irradiation, the PTFE grades MP1100 and MP1200 contain persistent perfluoroalkyl (peroxy) radicals, carbonyl fluoride end groups (–COF), and carboxylic acid end groups (–COOH), while non-irradiated MP1600 only has a very low proportion of carboxylic acid groups resulting of a special synthesis route. Due to the different types of polymerization procedures and irradiation MP1100, MP1200 and MP1600 show different particle sizes and particle size distributions. Fig. 1 demonstrates the particle size distributions of the utilized PTFE specifications, which were determined by dynamic light scattering measurements.

The different grades of PTFE were chemically coupled with the PA46 and PA66 matrix via reactive melt extrusion. The properties of used PA46 and PA66 are showed in Table 2, whereby the glass transition temperature of PA66 Technyl A205F is shown in [20].

The reaction is based on a transamidation reaction (see Fig. 2) as it was proved earlier for PA 6 [21].

The PTFE and PA were dosed in a ratio of PTFE/PA 20/80 wt-% and processed by Leistritz ZSE 27 Maxx twin-screw extruder (Leistritz Extrusionstechnik GmbH, Nuremberg, Germany). The maximum process temperature was 330 °C. In order to carry out the tribological tests, semi-finished products were prepared by injection molding of compounds (Allrounder 420 C, Arburg, Loßburg, Germany) at cylinder temperature of 290 °C (PA66) or 310 °C (PA46).

2.2. Differential scanning calorimetry (DSC)

The compounds were analyzed by a DSC device Q1000 (TA Instruments, New Castle, Delaware, USA) at a heating rate of 10 K/min between –30 °C and 350 °C under nitrogen atmosphere. The measuring cycle was heating, cooling and second heating. The temperature and heat of fusion were calibrated by indium (In) and lead (Pb) standards. The sample mass was 5–7 mg.

2.3. Tribological tests

The application of the PA-PTFE lubricant in worm gears is desired. Worm gears are usually made with a hardened steel worm and a bronze worm wheel. It is intended to transfer the dry lubricant from a sacrificial component via the worm into the gear contact. The loads investigated in this work to characterize the dry lubricants correspond to the load range that will be applied in real gears at the contact between the sacrificial component (dry lubricant) and the worm surface (steel).

The wear of compounds can be interpreted as a consumption of the dry lubricants. The higher the wear, the more material can theoretically be transferred to the steel counter body. This can be beneficial with regard to the goal of providing lubricant for the steel-bronze contact in the gear transmissions. Nevertheless, high wear of the PA-PTFE-cb compound leads to shorter service lives of the lubrication system, which in turn should be avoided. Regardless of where the optimum is in terms of wear, the prediction of wear is important for the design of the lubrication system.

The PA-PTFE-cb compounds were tribologically investigated after their processing into blocks (size: 70 × 70 × 15 mm³) via injection

Table 1
List of used PTFE type and their properties [36].

PTFE Type (Zonyl)	Type of Polymerization	Radiation Dose (kGy)	Surface Area (m ² /g)	Bulk Density (g/L)
MP1100	Emulsion	500	5–10	300
MP1200	Suspension	500	1.5–3	450
MP1600	Emulsion	0	8–12	350

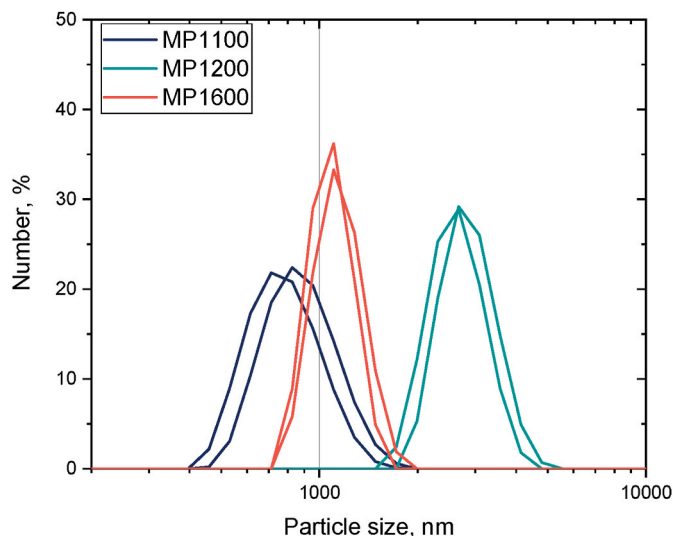


Fig. 1. Particle size distributions of the various PTFE types.

Table 2

List of used PA and their properties [34], [35].

	PA66 Technyl A205F	PA46 Stanyl TW300
Viscosity number (cm ³ /g)	natural	heat stabilized
Melting temperature (10 °C/min, °C)	263	295
Temp. of deflection under load 1.8 MPa (°C)	75	190
Temp. of deflection under load 0.45 MPa (°C)	220	280
Glass transition temperature (°C)	64–68	75

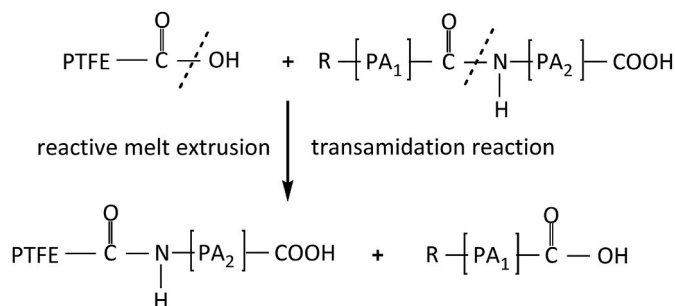
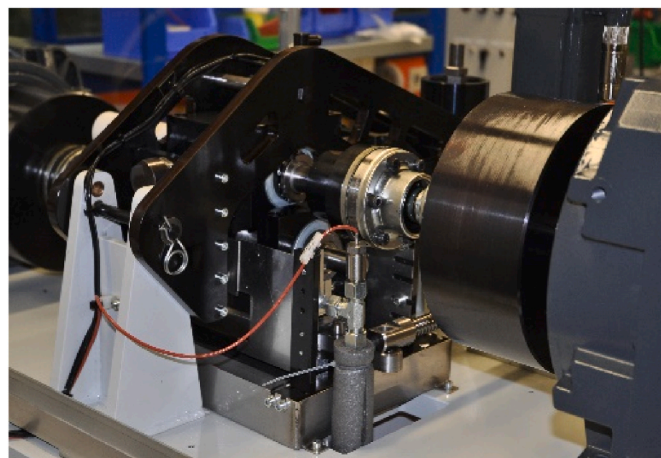
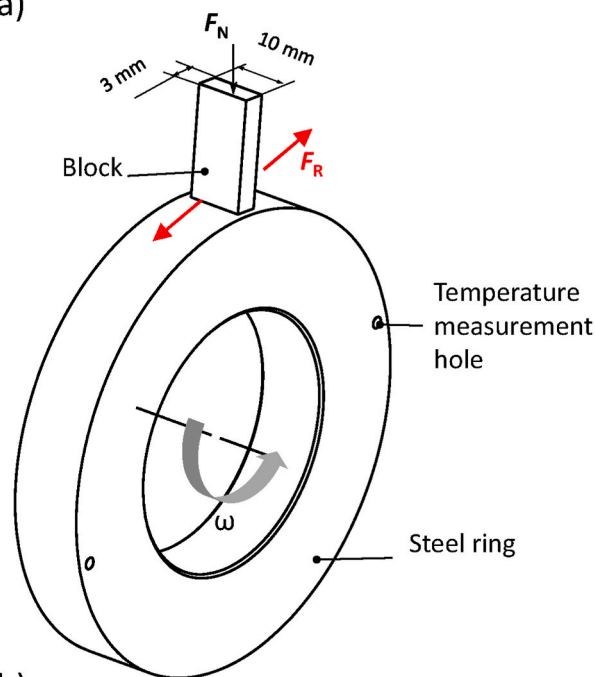


Fig. 2. Transamidation reaction between PA and radiation modified PTFE.

molding and subsequent milling. To better evaluate their tribological properties in contact with steel counter parts, comparative tests with their corresponding matrix materials, plain PA46 and PA66, were carried out. The tribological tests were performed on a block-on-ring test arrangement with similar contact and operating conditions compared to the corresponding gearbox application. The used test rig is described in [22]. Within the tests, wear of the compound material as well as the coefficient of friction between the steel ring and the compound block were determined via measuring the applied normal force F_N and the resulting frictional force F_R (see Fig. 3). The blocks were made of PA-PTFE-cb compounds and the ring of hardened (60 HRC) and transversely ground (with a roughness value of $Sa \approx 0.2 \mu\text{m}$ according to DIN EN ISO 25178) steel disc made of 16MnCr5. The PA-PTFE-cb compound had a cross-sectional area of $10 \times 3 \text{ mm}^2$, with a width of 10 mm and a length of 3 mm. The steel disc had a diameter of 80 mm and a width of 13 mm. A schematic view of the test set-up is shown in Fig. 3. In order to create a uniform pressure distribution between the PA-PTFE-cb compounds and the steel discs, the compounds were ground (with a



a)



b)

Fig. 3. a) MEGT's block-on-ring test rig, b) scheme of the test setup.

roughness value of $Sa \approx 1 \mu\text{m}$ according to DIN EN ISO 25178) in the working (rotation) direction to the radius of the steel counter sample. Before testing, the steel rings and the PA-PTFE-cb compounds were cleaned in an ultrasonic bath. The steel rings were cleaned using cyclohexane, isopropanol and acetone. Ethanol was used to clean the PA-PTFE-cb compound samples. After cleaning, the samples were dried under room conditions at 20–25 °C at least for 24 h. For wear measurement, the cleaned dry lubricant specimens are weighted individually before and after the test on a high-precision scale from Ohaus®, model Explorer EX225D with a precision of 0.01 mg. The temperature change of the steel discs was also recorded. For this purpose, a resistance thermometer was placed approx. 2 mm below the contact surface. Due to the deformation and wear of the compound, the temperature change in the compound was not measured here. The contact pressure was varied between 2 and 5 MPa, the sliding speed was kept constant at $0.5 \text{ m}\cdot\text{s}^{-1}$, and the test duration was 12 h, corresponding to a sliding distance of approx. 22 km. For each test configuration, three repeated measurements were carried out in order to determine the statistical certainty. The error bars in the results section of this paper represent the standard deviation of the measured variables. In this project it is

envisaged to carry out real gearbox tests under atmosphere operating conditions. Therefore, the tribological experiments presented here have been carried out under ambient conditions with fluctuating relative humidity between 25 and 50 %.

2.4. Surface analysis

Transfer films formed on the steel rings were analyzed by using a JEOL scanning electron microscope (JSM-6460 LV, Japan) adopted with an energy dispersive X-ray spectroscopy system in high vacuum mode (10^{-5} Pa) with an acceleration voltage of 20 kV. For the energy dispersive X-ray analyses a Genesis 7000 X-ray spectroscopy system from EDAX was used. An ALTURA 875 (FEI) DualBeam focused ion beam (FIB) system was utilized to prepare cross-sectional cuts through tribofilm layer on the steel surface. FIB milling was performed with a 30 keV gallium ion beam source. Before performing the FIB milling, a platinum layer was deposited on the sample in order to preserve the surface topography. After the FIB milling, SEM images of the cross-sectional cut and the surface around the cut were recorded.

The aim of surface analysis is to quantify the total amount of dry-lubricant material transferred onto steel surface. A straightforward method to do this is to measure directly the thickness of the material being transferred by using FIB (Focused Ion Beam). An alternative option is to measure the mass thickness as a suitable quantity, i.e. the total mass of lubricant covering a unit area. It can be determined from the attenuation of the EDX - Fe signal from the steel solids, which is caused by the overlying lubricant layer, depending on its local mass thickness. The latter has the advantage that relatively large surface areas can be measured, whereas the FIB method for measuring the layer thicknesses can only be carried out locally with reasonable effort. In this respect, it is less representative of the entire ring or disk surface covered with lubricant. These measurements permit the determination of transfer rates and correlations with characteristic values (e.g. COF, wear rate).

The transfer film thickness distribution of the dry lubricant compound, which forms in block-on-ring tests, is analyzed in the scanning electron microscope (SEM) by the energy-dispersive X-ray analysis (EDX) by means of element mapping over an area of 17.5 mm^2 . The Fe-intensity from the steel ring surface and the intensities of the spectral lines excited in the tribofilm are used as measurands. Essentially, the absorption of the X-rays emitted by the steel sample and its weakening by the tribofilm on the steel surface is evaluated. For this purpose, the intensities of Fe are measured once in a tribologically unstressed area of the steel surface and on the other hand on the resulting tribofilm. The

low-energy Fe $L\alpha$ spectrum with 0.705 keV was selected, the acceleration voltage of the electrons was 20 keV. Under these conditions, the strongest possible absorption of the characteristic radiation in the tribolayer volume was achieved with energetically favorable X-ray excitation. For the quantification of spectro-chemical data, information of X-ray absorption is required. Analytical descriptions of the coefficients of incidence used in the computer-aided evaluation of the spectra were derived from literature [23,24].

While the determination of the tribofilm thickness based on FIB-cuts would only be possible locally and on a few positions with reasonable effort, this method therefore allows a spatial-resolved determination of the tribofilm thickness over larger surface areas. The mass thickness of dry lubricant (MT) is determined by the product of the assumed density of PA-PTFE-cb compound density ($\rho_c = 80 \% \rho_{PA} + 20 \% \rho_{PTFE}$) and the determined layer thickness (dL), see equation (1).

$$MT = \rho_c \cdot d_L \text{ (g/cm}^2\text{)} \quad (1)$$

A comparison of the layer thicknesses measured in this way with corresponding measurements on FIB cuts, for the polymer PA66, shows that both methods deliver comparable results (Fig. 4). Backscatter electron image shows the lateral distribution of dry lubricant, which appears dark here due to the lower density to the Fe. FIB cuts were placed at 3 positions. It can be seen that the layer thickness is between 1.2 and $4.3 \mu\text{m}$. A comparison with the layer thicknesses measured by EDX at the same position, shows that this method is a good estimate of the tribofilm thickness on the steel surfaces.

3. Results and discussions

3.1. Results of DSC and SEM

After the melt extrusion, the chemically coupled PA46/PA66-PTFE-cb samples were examined using DSC, microscopic methods, and DLS as described earlier by Pompe et al. for PA6-PTFE-cb [21]. They have already found a correlation between fractional crystallization and the distribution quality of PTFE particles in the polyamide matrix. The fractional crystallization increases with a rising radiation dose of PTFE. This behavior reflected nicely the results from the microscopic methods and DLS. Dispersed PTFE in PA-PTFE-compounds crystallizes at the bulk crystallization temperature ($T_{c, \text{bulk}}$), which corresponds to the crystallization temperature of pure MP1100 (about 310°C) [21,32,33]. According to the particle size distribution, it is observed that the crystallization of finely dispersed PTFE with a small particle size occurs

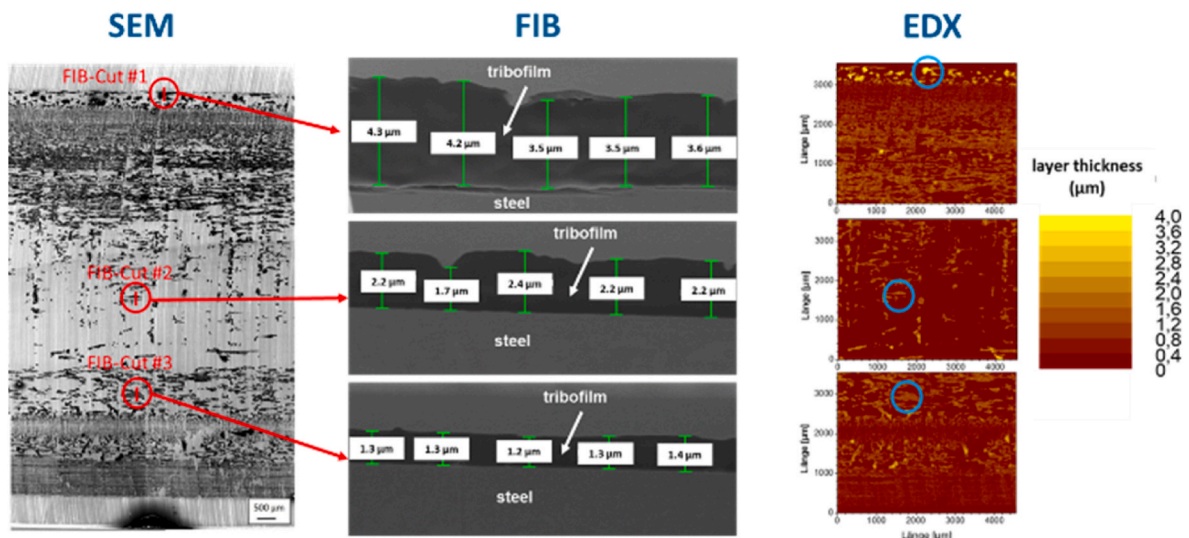


Fig. 4. Backscatter electron image visualizes the lateral distribution of PA66 tribofilm-thickness measured by FIB-Cut and by EDX.

at a lower temperature only. This is approximately 40 K below the $T_{c,bulk}$ and is described as fractional crystallization ($T_{c,frac}$ at about 270 °C). The higher the proportion of fractional crystallization, the better distribution of fine PTFE particles in the polyamide matrix.

In this paper, the crystallization behavior of the compounds was investigated by DSC. First, the compounds were heated up to 350 °C and held for 5 s. The crystallization behavior of the PA compounds was then determined. Since the melting behavior of PA and PA-PTFE compounds is very similar, heating curves were not used for interpretation here. Fig. 6 therefore only shows the cooling curves of the compounds shown in Fig. 5. During crystallization from a melt, crystallization takes place significantly below the melting temperature. The reason for this is the lack of a crystallization nucleus, which is required for crystallization.

As the crystallization curves of MP1100, MP1200 and MP1600 are very similar, it was decided to use the MP1100 crystallization curve as a model for PTFE crystallization curve in the following analyses (see Fig. 5). In comparison, the crystallization temperature of MP1200 is marginally higher than that of MP1100 and MP1600, which is also expected because the particle size of MP1200 is larger than that of MP1100 and MP1600 (see Fig. 1).

Fig. 6 shows the crystallization curves of plain PA46, original PTFE and compounds based on PA46, and the PTFE. PA46-MP1200-cb shows the smallest peak area of PTFE-bulk crystallization (at around 310 °C). Although the peak for fractional crystallization of the compounds is more or less overlapped by the crystallization peak of PA46, it can be concluded from the peak area differences in the bulk crystallization peak that PA46-MP1200-cb exhibits better compatibilization than the other PA46-PTFE samples obtained under comparable extrusion conditions. The compound with MP1600 shows the worst compatibility because it is not radiation modified and contains few polar coupling groups due to their synthesis, only. Therefore, it is structurally more similar to a physical blend between PA and PTFE. Consequently, splitting and fine distribution of PTFE into primary particles was also not anticipated [21].

The results from the SEM investigation confirmed this assumption (Fig. 7). The MP1200 particles are much more homogeneously distributed in the PA46 matrix than MP1100 and MP1600 and offered the best breaking down behavior in the range.

Using the SEM images, the particle size distribution can be determined by ImageJ software. The distribution of PTFE in PA46 is shown in the histogram in Fig. 8. MP1100 offers a broader distribution than MP1200, and the PTFE cut surface was determined up to 20 μm^2 , whereas for MP1200 up to about 4 μm^2 , only. This clearly means that the

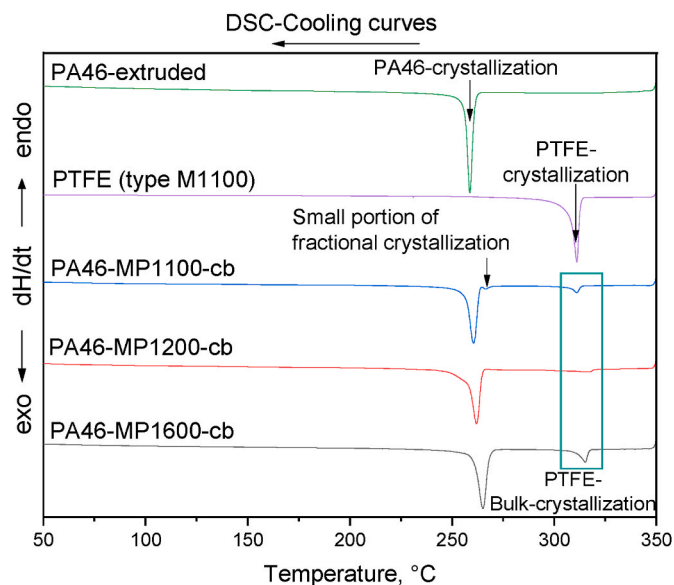


Fig. 6. DSC cooling curves of PA46-PTFE depending on the PTFE type.

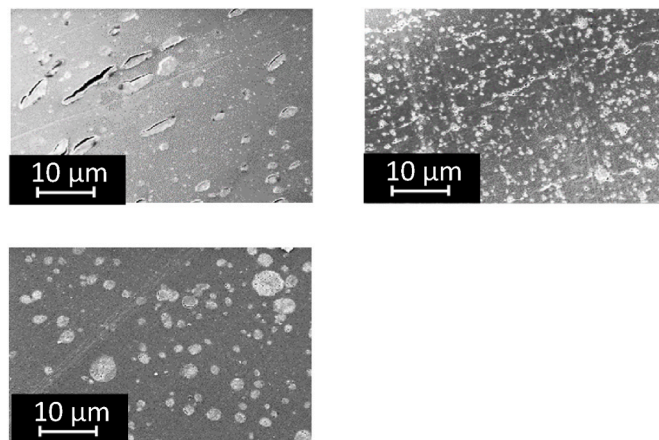


Fig. 7. SEM-Picture of PA46-PTFE-cb. a) top left: PA46-MP1100-cb; b) top right: PA46-MP1200-cb; c) bottom: PA46-MP1600-cb.

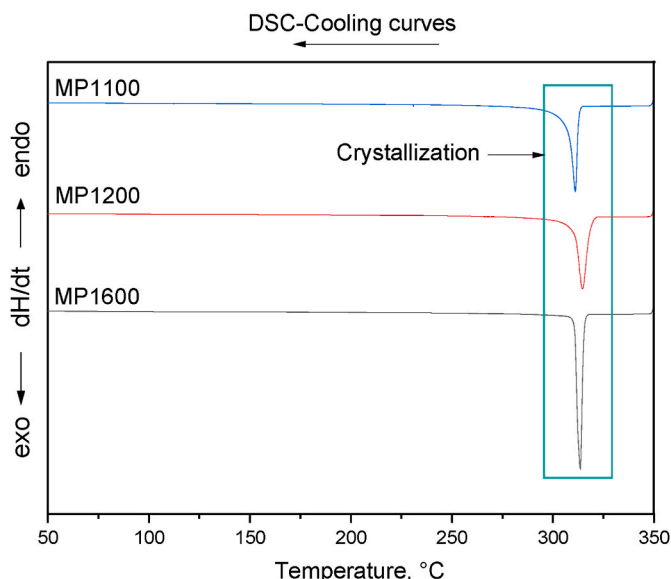


Fig. 5. DSC-cooling curves of used PTFE grades.

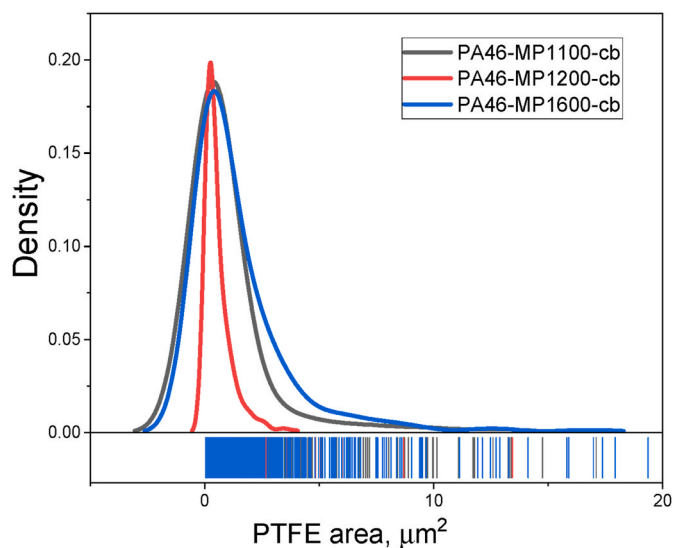


Fig. 8. Particle size distribution of PTFE in PA46 matrix.

chemical compatibility of PA46-MP1200-cb compounds is higher than PA46-MP1100-cb compounds.

In Fig. 9, DSC-cooling curves of plain PA66 and PA66-PTFE-cb compounds varieties are collected. Here, it can be seen that the sample PA66-MP1100-cb exhibits the most significant peak of fractionated crystallization at around 270 °C. It means that MP1100 is very well distributed in the PA66-matrix. The other two PA66-PTFE-cb samples do not show this fractionated crystallization. So, those PTFE particles' distribution and breaking down behavior are not as good as was observed for the PA66-MP1100-cb compound. However, it should be considered that the reactive extrusion process has not been optimized yet for each PA66/PA46-PTFE-cb combination. Moreover, the different melting behavior between PA66 and PA46 should be another factor that first PA-PTFE-cb compound MP 1100 shows the better compatibilization and higher melting PA46 MP1200.

The ImageJ software was also used for particle size analysis of PTFE in the PA66 matrix. Nevertheless, because it is clear from the SEM images that the particle size of MP1600 is enormous, this compound was not analyzed using ImageJ software. Fig. 11 shows the histogram of PA66-MP1100-cb and PA66-MP1200-cb compounds. The distribution of PTFE type MP1100 is much narrower than MP1200, and the cut surface of MP1100 particle was max. 7 μm^2 , in contrast, the cut surface of MP1200 particle was up to 20 μm^2 . Compared to PA46-MP1200-cb compound, there is a significantly broad distribution of MP1200 in PA66-matrix.

3.2. Friction and temperature in block-on-ring tests

The tribological characterization of various compound materials via wear measurement is accompanied by measurement of friction and temperature. While wear is only determined at the end of each test, friction and temperature can be recorded continuously.

Fig. 12 shows the characteristic curve of the coefficient of friction (COF) and the ring temperature θ_{Steel} as a function of the sliding distance at a pressure $p = 3 \text{ MPa}$ and a sliding velocity $u_s = 0.5 \text{ m/s}$. A comparison is made between the friction behavior of the plain PA66 and the

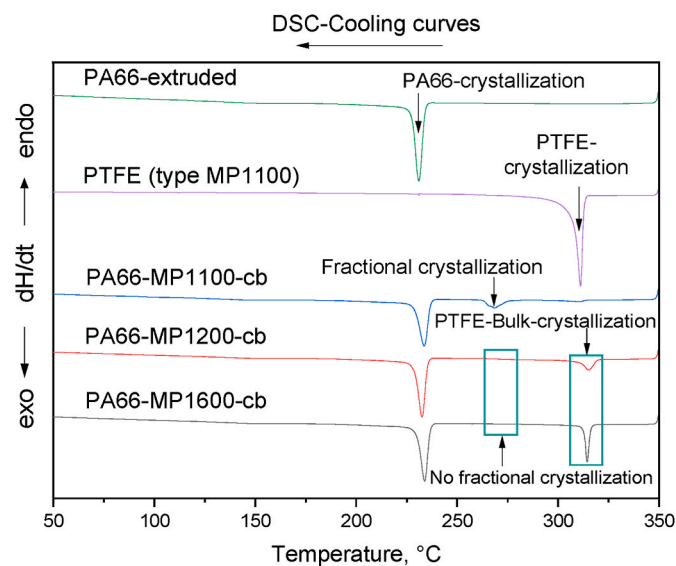


Fig. 9. DSC cooling curves of PA66-PTFE-cb depending on the PTFE type. SEM investigations of the PA66-PTFE-cb samples proved that MP1100 is very well disaggregated and distributed in the PA66 matrix (see Fig. 10a). Besides, according to the DSC curves, this compound offers the highest peak area of fractionated crystallization compared to PA66-MP1200-cb and PA66-MP1600-cb. In Fig. 10c, the MP1600 particles are comparatively enormous, and their distribution in PA66 is also very inhomogeneous. These results confirm those obtained by DSC investigations.

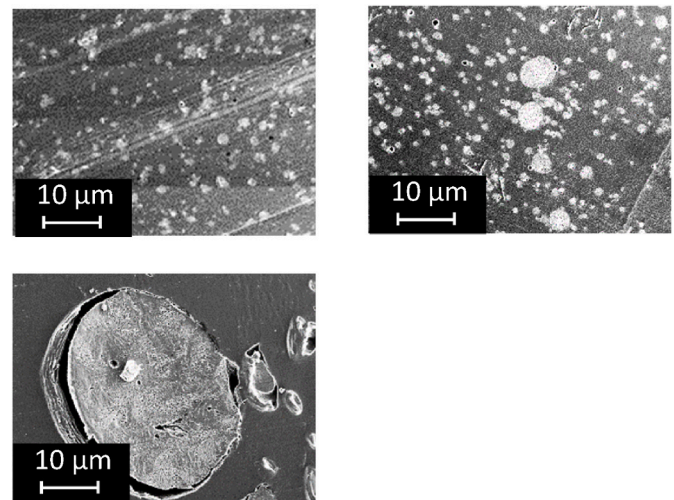


Fig. 10. SEM-Picture of PA66-PTFE-cb. a) top left: PA66-MP1100-cb; b) top right PA66-MP1200-cb; c) bottom PA66-MP1600-cb.

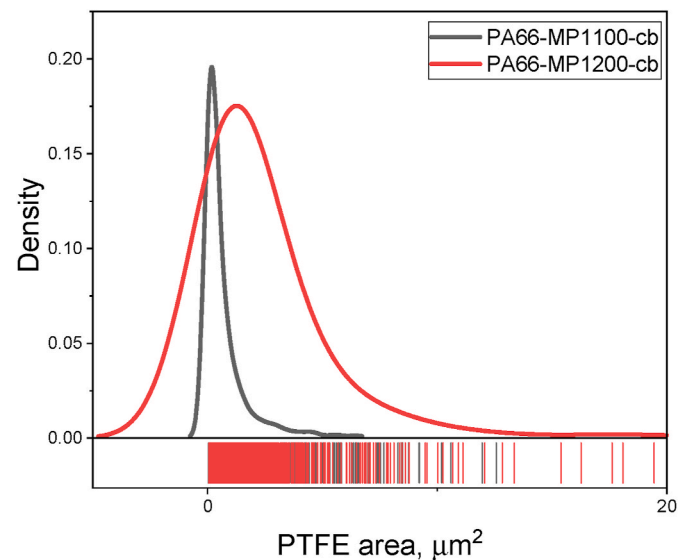


Fig. 11. Particle size distribution of PTFE in PA46 matrix.

PTFE-modified variant PA66-MP1100-cb. It should be noted that the temperature of the steel ring measured here is in reality lower than the real temperature in the contact zone. It can be seen in Fig. 12 that in the running-in phase, up to a sliding distance of approx. 2 km, the friction coefficient in the contact zone between the compound and the steel ring as well as the temperature of the ring increase rapidly. This running-in behavior can be observed with the plain PA66 and the PTFE-modified cb compound. Although the coefficient of friction for the two types of dry lubricants gradually drops to a nearly constant value after running-in, the temperature of the steel ring remains more or less constant when sliding with PA66. In contrast, the temperature of the steel ring drops by about 4 K when sliding with the compound PA66-MP1100-cb.

In addition, it can be observed how the chemical bonding of radiation-modified PTFE leads to a reduction in the coefficient of friction and the ring temperature in contact. The increase in friction coefficient and temperature in the run-in as well as the following decrease in friction coefficient have been investigated in various experimental works [27–29]. Pleskachevsky and Smurugov [28] investigated the influence of temperature on the frictional behavior of PTFE when sliding with a steel ring in a block-on-ring set-up. It was found that the drop in friction

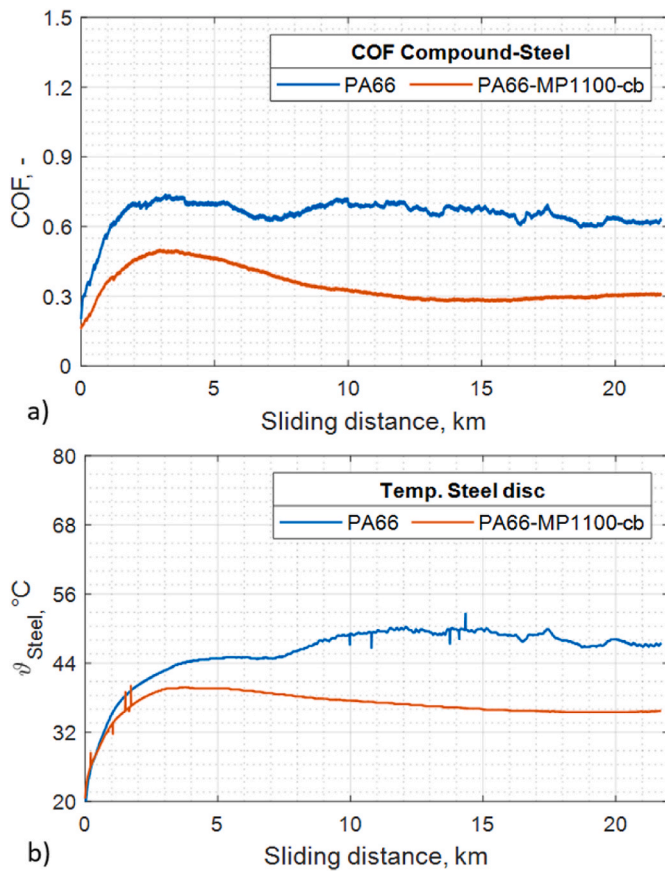


Fig. 12. Evolution of the coefficient of friction and ring temperature during the tribology process at a pressure $p = 3$ MPa and a sliding velocity $u_s = 0.5$ m/s.

coefficient was due to the transfer of the lubricant particles as well as the formation of a separating film on the metal surface, which prevented further heating of the steel ring. The behavior shown in Fig. 12 is typical for almost all compounds investigated in this work.

3.3. Compound wear in block-on-ring tests

The wear of the dry lubricants in use is of great importance, as it allows the amount of possibly transferred dry lubricant onto the steel disc to be estimated. In this work, the wear was determined gravimetrically, whereby the masses of the individual dry lubricants were measured before and after the sliding tests. Based on the experimentally determined densities of the dry lubricants, the volumetric wear is calculated according to equation (2) [].

$$\Delta V = \frac{\Delta m}{\rho} \quad (2)$$

where Δm describes the worn mass and ρ the density of the dry lubricant. The densities of the tested dry lubricants are listed in Table 3.

Table 3
Measured density of the tested dry lubricants.

Matrix	PTFE	Density /g•cm ⁻³
PA46	–	1.1805
	MP1100	1.2914
	MP1200	1.3011
	MP1600	1.3007
PA66	–	1.1355
	MP1100	1.2470
	MP1200	1.2555
	MP1600	1.2580

The results from the density measurement show that, on the one hand, the chemical bonding of irradiated PTFE to PA46 or PA66 leads to an increase in density of the compounds. On the other hand, the different PTFE types do not lead to any significant density changes of the compounds.

The comparison of the volume wear in Fig. 13 shows that with increasing pressure the wear of the matrix polymers PA46 and PA66 increases, whereby the polymer PA66 reaches higher values.

Fig. 14 shows the influence of the contact pressure on the steel ring temperature at the end of the sliding contact between the plain PA and the steel ring, whereby an increase in the temperature with increasing pressure is observed. The temperatures are compared according to the quasi-stationary range of the characteristic curves, from a sliding distance of approx. 15 km, see Fig. 12. For this purpose, an average value of the temperature is calculated within this section.

Fig. 15 shows the volumetric wear of the modified dry lubricants, whereby the influence of the different PTFE types on the polymer matrices and the pressure on the wear behavior is shown. In comparison to the wear of plain polymers shown in Fig. 13, it can be observed that, the bonding of radiation-modified PTFE into the PA46 and PA66 matrices considerably reduces the volume wear, independent of the PTFE type and the polymer matrix. A similar behavior can be observed with the temperature of the steel ring, whereby the steel ring heats up more when plain PA matrices materials are used as counter bodies to the steel ring.

Fig. 16 shows the influence of irradiated PTFE on the ring temperature when the PA-PTFE-cb compounds are sliding onto the steel ring. Compared to the temperature of the steel ring when the plain PA matrices are used, it is obvious that the bonding of radiation-modified PTFE into the matrices PA46 and PA66 leads to a decrease of the temperature of the steel ring. It can also be observed that with increasing pressure, the temperature of the steel ring becomes higher and thus it can be assumed that the temperature in the contact area of the friction partners also increases.

In general, the tribological behavior of polymers can be related to their viscoelastic and temperature-dependent properties. The sliding motion between two materials leads to an increase in temperature at the contacting surfaces of the two materials, which influences the viscoelastic properties, shear forces, adhesion and transfer behavior.

The increase in wear volume (see Fig. 13) of PA66 compared to PA46 may be explained by the fact that the onset glass transition temperature T_g of PA66 ($T_{g \text{ Onset}} = 46$ °C) is significantly lower than for PA46 ($T_{g \text{ Onset}} = 58$ °C), as determined by DSC measurements (Fig. 17). Therefore, the mechanical properties of PA66 may change at an earlier test stage than PA46. The temperature increase in the tribotest provokes increased wear. In contrast, T_g was determined by the peak maximum of $\tan \delta$ in the dynamic mechanical analysis (DMA) at nearly identical

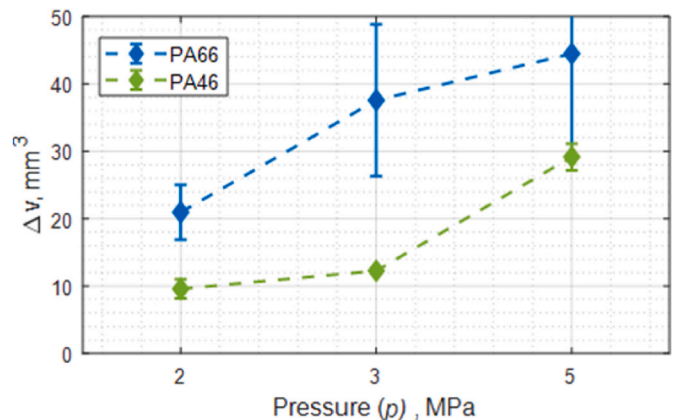


Fig. 13. Influence of the contact pressure on the wear of the polymers PA46 and PA66.

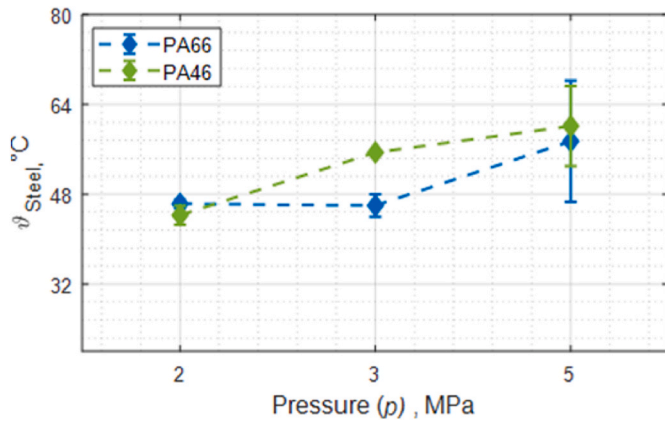


Fig. 14. Influence of the contact pressure on the steel ring temperature when the plain polymer matrices PA46 and PA66 are used.

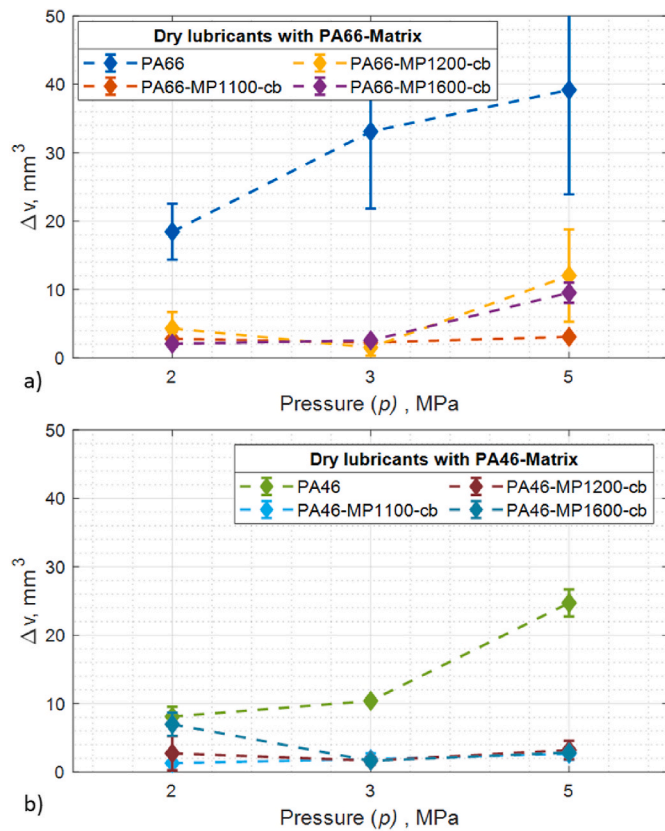


Fig. 15. Comparison of the wear volumes for the individual PA-PTFE-cb compounds with different polymer matrices depending on the contact pressure.

temperatures.

The improved wear resistance of the compounds containing irradiated PTFE (see Fig. 15) can be attributed to the irradiated PTFE on the contact surface, whereby the self-lubricating effect of PTFE lubricant can reduce the molecular chain adhesion in the polymer matrices [16]. The ring temperature during the test was approximately 65 °C when the plain PA46 and PA66 was used, whereas it decreased to approximately 45 °C when the compounds PA-PTFE-cb with 20 wt-% irradiated PTFE were analyzed. This suggests that the improved wear resistance of the compounds also relied on the lower temperature at the sliding surface.

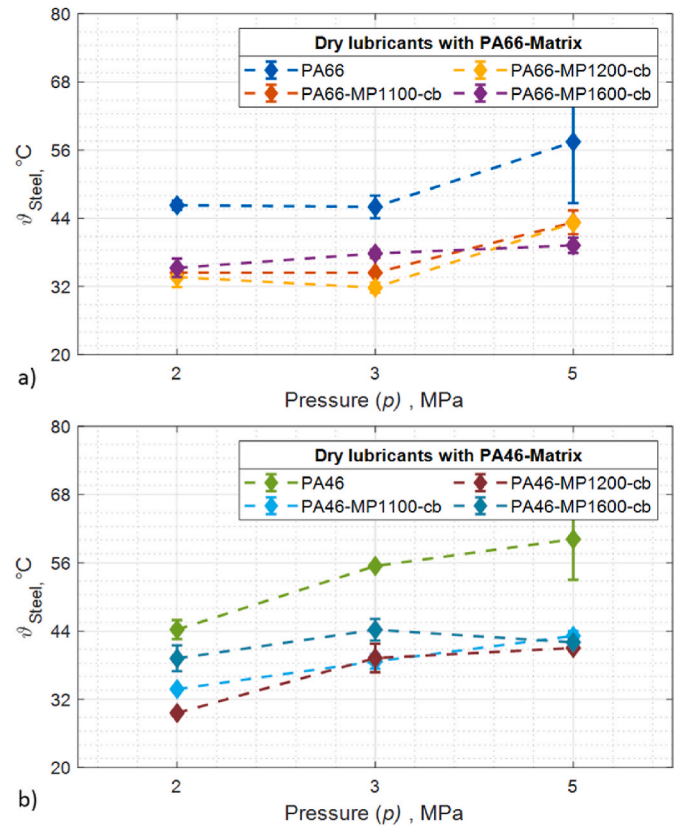


Fig. 16. Influence of the contact pressure and the polymer matrices on the steel ring temperature when the compounds with irradiated PTFE are used.

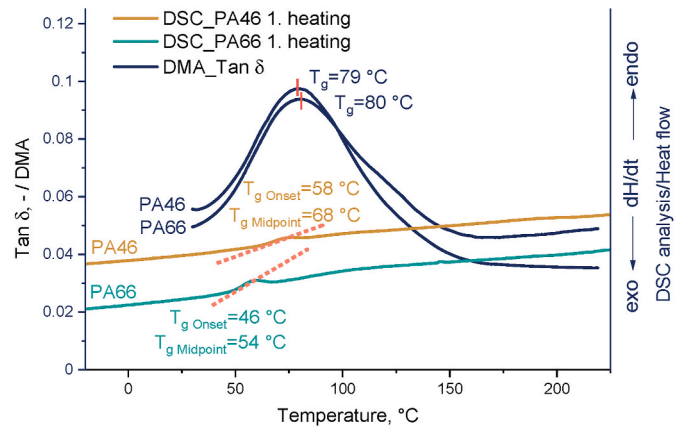


Fig. 17. DMA and DSC analyses on used PA type to determine the T_g temperatures.

3.4. Comparison of different compounds regarding friction in block-on-ring tests

Fig. 18 shows the dependence of the friction coefficient on the pressure in the contact area of polymer PA46 and PA66 towards the steel ring, where the coefficient of friction of the polymer PA66 is relatively lower than that of the polymer PA46. As well as the temperature of the steel ring, the coefficient of friction is compared here according to the quasi-stationary range of the characteristic curves, from a sliding distance of approx. 15 km, see Fig. 12. Whereby an average value of the friction coefficient is calculated over all measured values within this section.

Various authors found that the coefficient of friction between

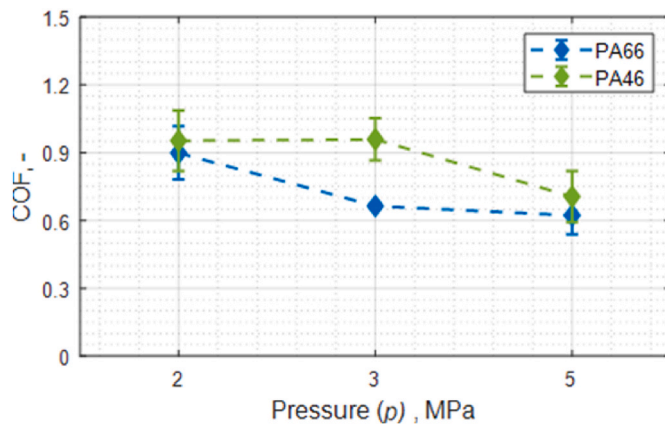


Fig. 18. Influence of the contact pressure on the coefficient of friction when using the polymer matrices PA46 and PA66.

polyamides and steel surfaces decreases with increasing temperature and load [5,30], due to the combined dependence of the modulus of elasticity and the shear force on temperature. Comparable frictional behavior was observed in this work for the plain polymers PA46 and PA66, see Fig. 18. In general, the coefficient of friction between polymers and steel surfaces is influenced by the adhesion of the polymer to the metal surface and the deformation of the polymer. An increase in temperature above the glass transition temperature of the polymers leads to a decrease in the Young's modulus and the shear strength of the polymers. The reduction of the Young's modulus leads to an increase of the real contact area between polymer and steel surfaces, as the roughness asperities of the steel surface penetrate the polymer more deeply. On the other hand, the decrease of the shear strength results in an easier deformation of the polymer. However, at higher temperatures the friction forces are dominated by the shear strength [5]. Fig. 15 shows that in contact with the steel disc, the PA66 samples show higher wear than the PA46 samples. As a result, compared to PA46, an increased amount of PA66 polymer could be deposited onto the steel surface, which could promote the formation of a transfer film of PA66 onto the steel surface. This could explain why, compared to PA46, PA66 leads to lower coefficients of friction in contact with the steel surface. Furthermore, it is described in [31] that a higher coefficient of friction of PA46 compared to PA66 is also due to the higher crystallinity of PA46 (60–70 %) compared to PA66 (~50 %).

The influence of different irradiated PTFE types on the friction behavior of PA-PTFE-cb compounds is shown in Fig. 19. As demonstrated before, the examined PTFE differ, among other things, in their degree of functionalization and the method of polymerization.

Comparing the results from Figs. 18 and 19, it can be seen that the coefficients of friction of the polymer matrices PA46 and PA66 are larger than the coefficients of friction of compounds modified with the different PTFE types (MP1100, MP1200 and MP1600). Furthermore, it is obvious from Fig. 19 that the coefficient of friction decreases with increasing contact pressure, when polymer matrix PA66 is used. Moreover, it can be assumed, that at relatively low pressures, the compounds containing PTFE grade MP1100 show the lowest coefficient of friction in contact with the steel ring. The decrease of the coefficients of friction using the PTFE-modified compounds can be attributed to the change in the mechanical and thermal properties of the solid lubricants due to the chemical coupling of PTFE in the PA-Matrices, which also influences the deformation and adhesion components of the friction. Furthermore, the different frictional properties between the individual dry lubricants can be attributed to their respective chemical compatibility with the steel surface and to the distribution of the PTFE dry lubricant in the polymer matrices, see Figs. 7 and 10. This difference in chemical compatibility is also influencing the amount of PTFE dry lubricant transferred to the steel surface and thus the self-lubricating effect of the dry lubricant,

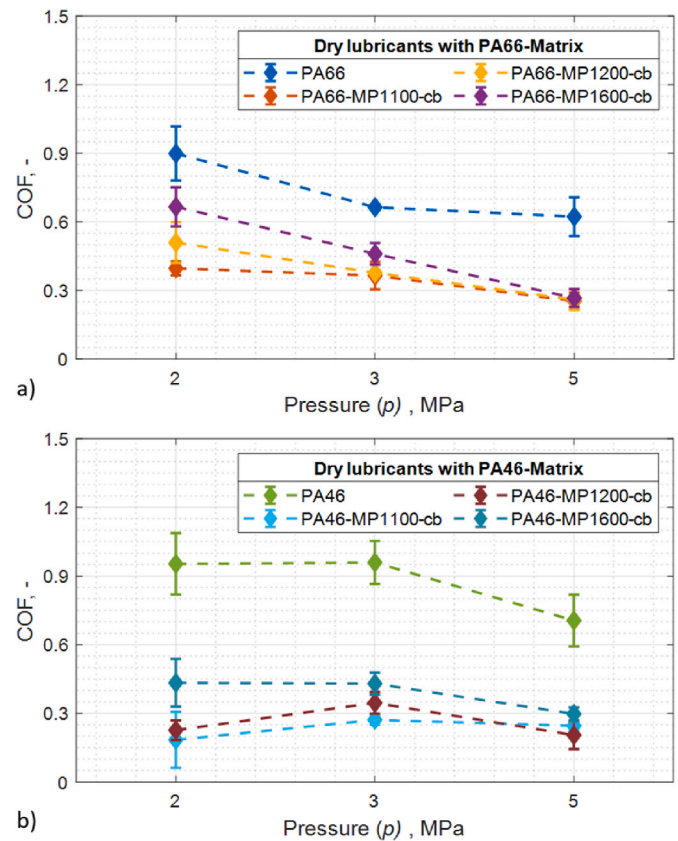


Fig. 19. Comparison of the coefficient of friction for the individual PA-PTFE-cb compounds with different polymer matrices depending on the contact pressure.

which also influences the wear behavior of the dry lubricants.

3.5. Surface characterization

After the tests on the block-on-ring test bench, the lateral distribution of the PA46 compound materials with the radiation-modified PTFE types MP1100 and MP1600 was visualized on the steel disc surfaces by means of light- and scanning electron microscopy (REM), as well as the resulting mass thickness by means of energy-dispersive X-ray analysis (EDX). Regardless of the respective composition and the contact pressure of the block on the steel disc, local, strip-shaped compound material accumulations are observed in the circumferential direction. Furthermore, components of the abrasively worn PA46-PTFE block material in the score marks can be detected (Fig. 20 and 21). Overall, an inhomogeneous tribo layer forms on the steel surface over the disc circumference and layer thickness.

By increasing the contact pressure, the gravimetrically determined total wear of the block increases significantly regardless of the PTFE type used. In contrast, however, the masses of the polymeric wear components deposited on the steel disc surface remain largely the same in the pressure variation, also independent of the PTFE type used.

As a result, with increasing block wear due to higher contact pressures, the proportion of compound mass remaining on the steel surface after the end of the tribo experiment is reduced due to higher contact pressures, see Fig. 22. The higher wear on the block is therefore not coined with a corresponding increase in the amount of compound material transferred to the steel surface. The proportion of the PA46-PTFE-cb polymer compound deposited on the steel surface, which is related to the overall wear of the block, decreases at higher pressure, in other words: the efficiency of the tribolayer build-up decreases.

On the other hand, the absorption capacity of the steel surface for polymeric abrasion, which is decisively determined by the specific

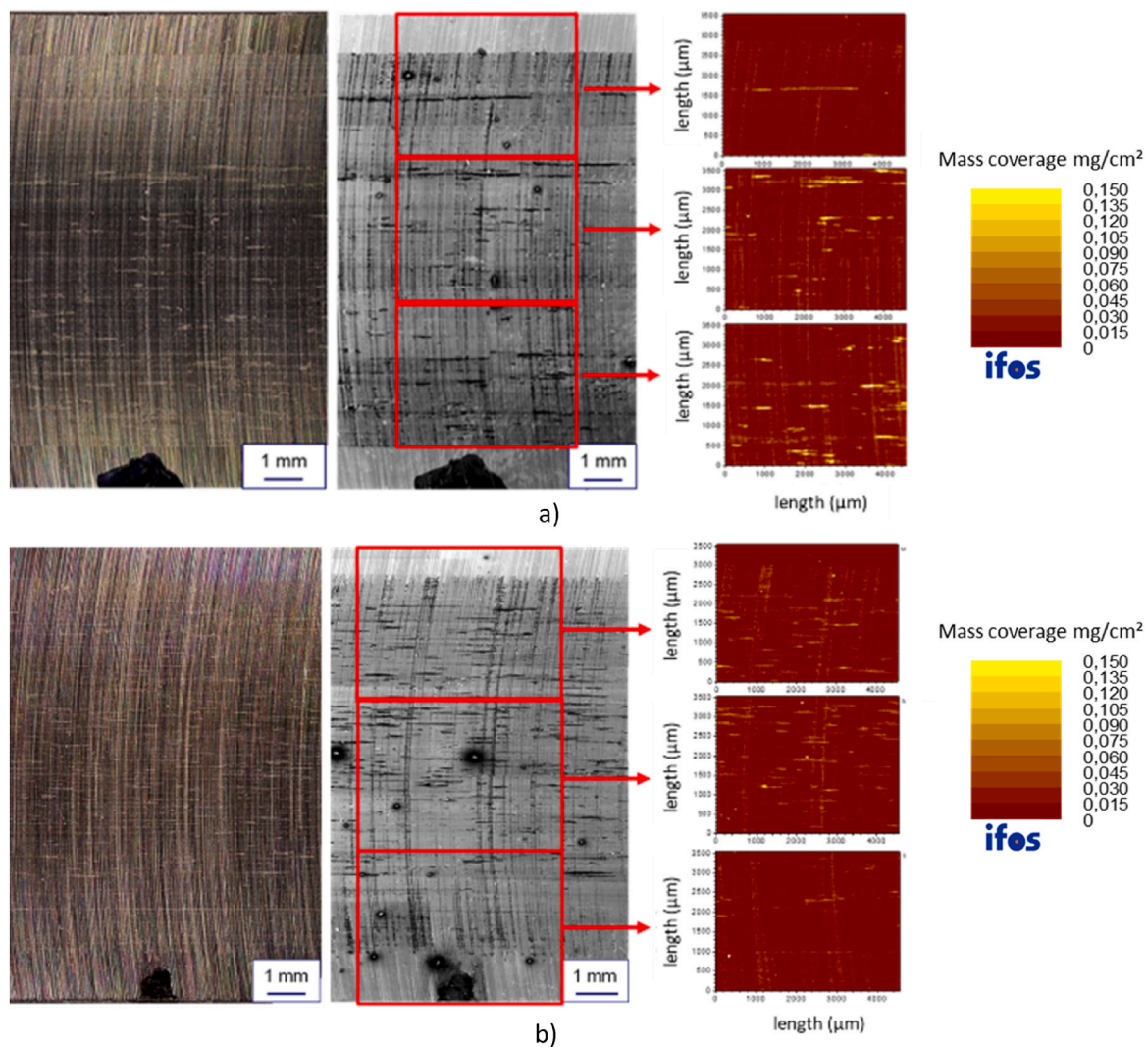


Fig. 20. Light microscopy (left), backscattered electron images (center) and EDX-measured distribution images of compound mass thickness (right) on representative steel surface sectors depending on the contact pressure of the P46-MP1100-cb compound selected in the tribo experiment. a) for 3 MPa and b) for 5 MPa.

topography, is limited in the sense of landfilling of dry lubricant. If the disc surface sanded transversely to the running direction is “saturated” with the material rubbed off on the block (e.g. the roughness depths or grooves filled), less and less wear material can be applied to the steel surface under the given tribological stress. The polymeric abrasion continuously produced in the contact block /ring then remains unused at least for a further assembly or removal of a triboinduced dry lubricating layer.

4. Conclusions

In this work, tribological and surface analytical investigations for the characterization of PA-PTFE-cb dry lubricants were performed, which were produced by reactive extrusion of radiation modified PTFE with PA46 and PA66 in the melt. The DSC measurements demonstrated that the various PA-PTFE compounds exhibit different compatibility and thus coupling qualities for the same extrusion parameters. Almost no bulk crystallization was detected for PA66-MP1100-cb and PA46-MP1200-cb, which correlates with a large amount of fractionated crystallization subsequent to reactive extrusion. From that behavior and SEM images a high level of compatibilization and thus chemical bonding between both polymers in those compounds can be concluded. Furthermore, according to the previous studies, the DSC results agree with the results of FTIR measurement. The compounds with a large

amount of fractionated crystallization also have a high degree of coupling between PTFE and PA, which has been proven by FTIR [26].

The polymer matrices PA46 and PA66 and the various PA compounds modified with PTFE were tested using a block-on-ring arrangement with steel counter samples at relatively low pressures. It was shown that, independently of the load, the PA-PTFE-cb compounds exhibited a lower coefficient of friction and higher wear resistance compared to the plain PA46 and PA66. The reduction in friction and wear of the compounds can be traced back to the chemical covalent coupling (PA-PTFE) of the compounds and their respective chemical compatibility with the steel surface. The higher the PA-PTFE coupling degree, the better the distribution of PTFE-micro particles in the PA matrix. Due to the lack of chemical covalent bonding between PA and PTFE, considering PA46-MP1600-cb and PA66-MP1600-cb compounds, there is no good distribution of MP1600 micro particles in the PA matrix, which could be a reason for the unfavorable tribological properties of compounds based on PTFE of grade MP1600.

Surface analytical studies on steel disc surfaces, which were previously tribologically stressed by PA46 compounds with differently radiation-modified PTFE fillers, showed that an increase of pressure in the block/ring-contact does not lead to an increase of the amount of dry lubricant material on the steel surface. The resulting tribofilms are very inhomogeneous even in terms of their lateral distribution or layer thickness. The topography of the ring surface determines the absorption

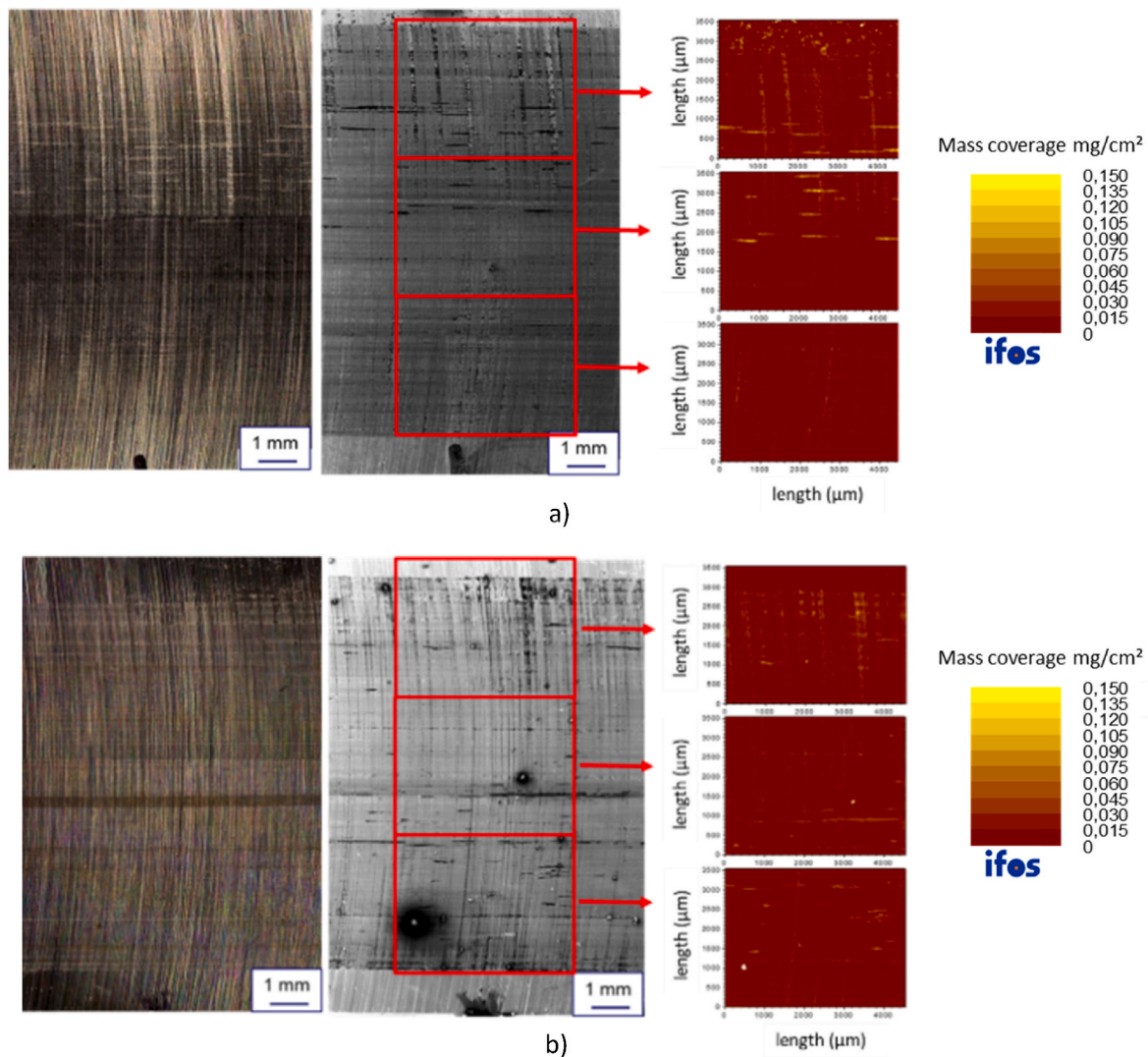


Fig. 21. Light microscopy (left), backscattered electron images (center) and EDX-measured distribution images of compound mass thickness (right) on representative steel surface sectors depending on the contact pressure of the P46-MP1600-cb block selected in the tribo experiment. a) for 3 MPa and b) for 5 MPa.

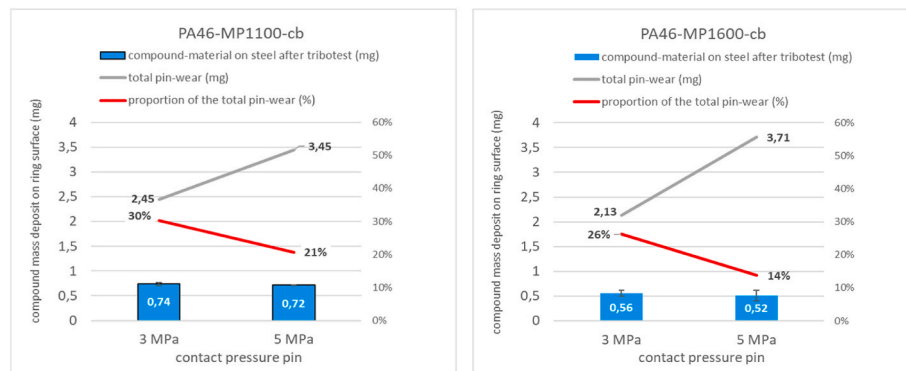


Fig. 22. Relationship between total compound wear and the amount of material deposited on the surface depending on the contact pressure.

capacity for polymeric abrasion components. With a pressure of 3 MPa, significantly less compound wear is measured compared to higher pressure (5 MPa). At the same time, the amount of mass thickness on the steel surface changes only slightly. This suggests that a more efficient lubrication mechanism can be achieved at low pressures. Surface analysis shows, that the use of dry lubricant polymer compound PA46-

MP1100-cb leads to a higher mass thickness on the steel surfaces than PA46-MP1600-cb, thus reducing the direct contact between the compound PA46-MP1100-cb and the steel surface. This could also explain the favorable friction behavior of the compounds based on PTFE grade MP1100, compared to the compounds with the PTFE grade MP1600. Furthermore, the reason for the higher mass of the compound PA46-

MP1100-cb on the steel surface could be the presence of the higher polarity of the -COOH-functional groups of the radiation-modified MP1100. And at the same time, through the chemical covalent bonding of PTFE to polyamide, PA segments could be found on the steel surface. Samples containing MP1600 have very low chemical bonding between polyamide and the PTFE. That is why wear tests result in pure MP1600 powder in their case. In other words, the lack of polar COOH functional groups in MP1600 leads to lower mass of compounds being deposit on the steel surface compared to compounds containing the radiation modified PTFE. These assumptions have to be verified on optimized compounds and are objects of future research. Furthermore, the influence of the built up tribofilm on the steel surface will be investigated in a steel/bronze contact.

Author statement

Lionel Simo Kanga: conceptualization, methodology, investigation, validation, writing - original draft, writing - review & editing, visualization, formal analysis, **Thanh-Duong Nguyen:** conceptualization, methodology, investigation, validation, writing - original draft, writing - review & editing, visualization, formal analysis, **Stefan Emrich:** conceptualization, methodology, investigation, validation, writing - original draft, writing - review & editing, visualization, formal analysis, **Thomas Schmidt:** investigation, **Manuel Oehler:** supervision, writing - review & editing, **Michaela Gedan-Smolka:** supervision, writing - review & editing, project administration, funding acquisition, **Michael Kopnarski:** supervision, project administration, funding acquisition, **Bernd Sauer:** supervision, project administration, funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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